A Universal Deep Learning Model for Ultrasound Segmentation and Classification

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Abstract. In this study, we developed a unified deep learning-based method for segmentation and prediction. The model shared one encoder with a different decoder for different tasks. The main innovation of this model is that we integrated different tasks with one shared model and without training the tasks individually.

Keywords: Ultrasound, Segmentation, Classification, Deep Learning

1 Introduction

2 Introduction

Ultrasound (US) imaging is one of the most widely used modalities in clinical practice due to its real-time acquisition, non-ionizing nature, and cost-effectiveness. It plays a crucial role in a variety of diagnostic and interventional procedures, ranging from breast cancer screening and thyroid nodule assessment to fetal anomaly detection and cardiac function evaluation. However, the interpretation of ultrasound images remains highly operator-dependent and is challenged by inherent limitations such as speckle noise, low contrast, and variability in anatomical appearance across patients and scanning devices. These factors often hinder both the accuracy and consistency of clinical decision-making.

Recent advances in deep learning have demonstrated remarkable success in automated medical image analysis, particularly in segmentation and classification tasks. Segmentation enables precise delineation of anatomical structures or pathological regions, which is essential for quantitative assessment and treatment planning. Classification supports diagnostic decision-making by identifying disease categories or stages directly from image data. While many studies have proposed specialized deep learning architectures for specific organs or tasks, these models typically lack generalizability, requiring separate training for each organ system or diagnostic objective. This task-specific nature limits their scalability and clinical deployment in multi-organ, multi-task scenarios.

To address these limitations, we propose a *universal deep learning model* capable of performing both segmentation and classification across diverse ultrasound tasks within a single unified framework. Our approach is motivated by the observation that different ultrasound analysis tasks share common low- and mid-level visual patterns, while differing in their high-level task-specific representations. By leveraging a shared residual encoder to extract generic multi-scale features and incorporating task-specific residual decoders and multi-scale classification heads, our model achieves efficient feature sharing while preserving task adaptability. Furthermore, we introduce a prompt-based conditioning mechanism to explicitly encode task identity and organ context, enabling the model to dynamically modulate its feature extraction and prediction strategy according to the task at hand.

The proposed architecture supports simultaneous training of segmentation and classification tasks using a dynamic task-weighted loss function, allowing the model to balance heterogeneous objectives without extensive manual tuning. Extensive experiments on multi-organ ultrasound datasets demonstrate that our framework achieves competitive or superior performance compared to specialized single-task models, while offering the advantages of reduced model storage, streamlined deployment, and enhanced generalization across tasks. This work represents a step toward a clinically viable, scalable, and adaptable AI system for comprehensive ultrasound image analysis.

Thus, our main contributions are as follows.

- We propose a straightforward and efficient network model for universal ultrasound classification and segmentation tasks.
- To improve the performance, the proposed network integrates Feature-wise Linear Modulation and prompts to adjust the specific tasks, which can reduce the model size.
- We extensively evaluated on the public and dataset shows the superior performance compared to single model segmentation and classification.

2.1 Related Work

Deep Learning for Ultrasound Segmentation. U-Net [10] and U-Net++ [14], have been widely adopted for medical image segmentation, including applications to breast [11], thyroid [4], and fetal ultrasound [1]. Despite strong performance, these architectures are often trained separately for each organ or modality, resulting in redundant models and limited scalability.

Deep Learning for Ultrasound Classification. Convolutional neural networks (CNNs) have been successfully applied to US-based disease classification, such as breast lesion malignancy prediction [8], pancreas [7] etc. More recent works explore attention mechanisms [7] and transformer-based architectures [3] to capture global context. However, these models remain task-specific and rarely share representations with segmentation networks.

Multi-task Learning in Medical Imaging. Multi-task learning (MTL) has emerged as an effective strategy for improving generalization by jointly learning related tasks [12]. In medical imaging, MTL frameworks have been proposed to combine

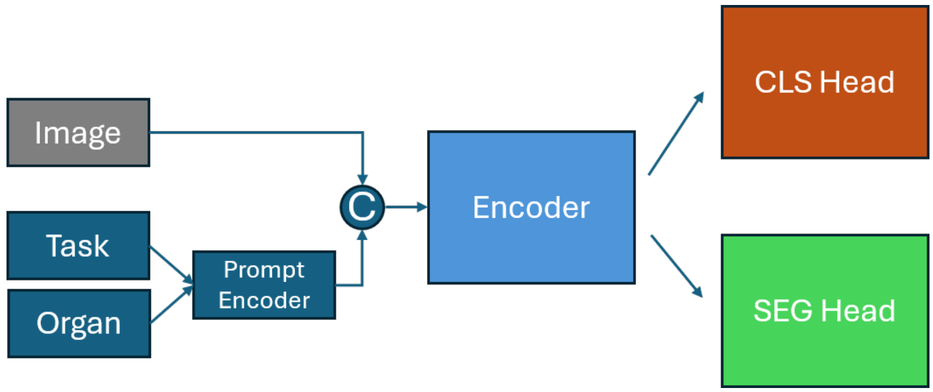

Fig. 1. The overall framework for ultrasound classification and segmentation .

segmentation and classification [2], leveraging shared encoders with task-specific heads. Nevertheless, most MTL approaches in ultrasound focus on a single organ, and their ability to generalize across diverse anatomical regions remains underexplored.

Prompt-based and Conditional Deep Learning. Prompting and conditional networks have recently been explored in vision-language models [5] and cross-domain adaptation [13]. In medical imaging, conditional mechanisms allow dynamic adaptation of feature extraction based on auxiliary inputs. Our work integrates a prompt-based task encoding into a residual MTL framework, enabling universal ultrasound analysis without retraining for new organs or tasks.

3 Methodology

As shown in Figure 1, the proposed method consists of two main parts: a shared encoder and task-specific decoder heads. The unified multi-task ultrasound analysis network that integrates both segmentation and classification within a shared encoder–decoder framework. The network adopts a residual learning paradigm in both the encoder and task-specific heads to enhance feature reuse and gradient flow.

3.1 Shared Encoder

For the shared encoder, we used ResNet152 as the backbone, which is stable and widely used in different computer vision tasks.

The encoder is based on a ResNet-152 backbone pre-trained on ImageNet, modified to accept $C_{\text{in}} \in \{3, 4\}$ input channels depending on whether task prompts are used. The encoder produces four hierarchical feature maps:

$\{\mathbf{F}_1 \in \mathbb{R}^{B\times 256\times H/4\times W/4},\ \mathbf{F}_2 \in \mathbb{R}^{B\times 512\times H/8\times W/8},\ \mathbf{F}_3 \in \mathbb{R}^{B\times 1024\times H/16\times W/16},\ \mathbf{F}_4 \in \mathbb{R}^{B\times 2048\times H/32\times W/32}\}.$

These features are shared across both tasks.

Encoder Task Prompt Integration When prompt conditioning is enabled, a learned task- and organ-specific prompt $\mathbf{P} \in \mathbb{R}^{B\times 1\times H\times W}$ is concatenated to the image input along the channel dimension, allowing the encoder to adaptively modulate feature extraction based on task identity.

To adapt the specific tasks, we proposed to use the Feature-wise Linear Modulation (FiLM) [9] for the task conditioning module.

FiLM is a conditioning method that adaptively influences the intermediate feature representations of a neural network by applying feature-wise affine transformations. Given an input feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$, FiLM applies a scaling and shifting operation to each feature channel:

$$\mathrm{FiLM}(\mathbf{F} \mid \boldsymbol{\gamma}, \boldsymbol{\beta}) = \boldsymbol{\gamma} \odot \mathbf{F} + \boldsymbol{\beta}, \tag{1}$$

where $\boldsymbol{\gamma}, \boldsymbol{\beta} \in \mathbb{R}^C$ are the modulation parameters, and $\odot$ denotes element-wise multiplication. These parameters are typically generated by a conditioning network based on an auxiliary input (e.g., a question in visual question answering or metadata in medical imaging).

FiLM enables the model to dynamically adjust feature processing according to the contextual information provided by the conditioning input. Unlike fixed normalization or attention mechanisms, FiLM directly modifies the activation values at the feature-channel level, allowing flexible and interpretable modulation of neural network computations.

3.2 Task Specific Decoder Head

Classification Head The segmentation branch employs a residual U-Net decoder. Each decoding stage consists of an upsampling block followed by a *Residual Double Convolution* (RDC) block:

$$\mathrm{RDC}(\mathbf{x}) = \sigma(\mathrm{BN}_2\left(\mathrm{Conv}_{3\times 3}(\sigma(\mathrm{BN}_1(\mathrm{Conv}_{3\times 3}(\mathbf{x}))))\right) + \mathbf{s}(\mathbf{x})),$$

where σ is the ReLU activation, BN denotes batch normalization, and $\mathbf{s}(\cdot)$ is a 1×1 skip projection when channel dimensions differ. The decoder progressively fuses encoder features via skip connections:

$$\mathbf{U}_i = \mathrm{RDC}([\mathrm{Up}(\mathbf{U}_{i+1}), \mathbf{F}_i]),$$

with $\mathrm{Up}(\cdot)$ denoting transposed convolution upsampling and $[\cdot,\cdot]$ channel-wise concatenation. A final RDC block followed by a 1×1 convolution outputs the segmentation map $\hat{\mathbf{Y}}_{\mathrm{seg}} \in \mathbb{R}^{B\times C_{\mathrm{seg}}\times H\times W}$.

Segmentation Head The classification branch employs a Multi-Scale Residual Classification Head (MS-RCH) to exploit hierarchical features. First, each encoder feature map $\mathbf{F}_i$ is projected to a common embedding dimension d_{emb} using a 1×1 convolution, batch normalization, and ReLU activation, then globally average pooled:

$$\mathbf{z}_i = \text{GAP}(\sigma(\text{BN}(\text{Conv}_{1\times1}(\mathbf{F}_i)))) \in \mathbb{R}^{B \times d_{\text{emb}}}.$$

The pooled features are concatenated to form $\mathbf{z} = [\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3, \mathbf{z}_4] \in \mathbb{R}^{B \times 4d_{\text{emb}}}$, and passed through a residual MLP:

$$\hat{\mathbf{y}}_{\text{cls}} = \mathbf{W}_2\, \sigma(\mathbf{W}_1\, \mathbf{z}) + \mathbf{S}\, \mathbf{z},$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ are learnable fully connected layers, and $\mathbf{S}$ is a linear skip projection matching dimensions.

3.3 Loss function:

Each task has a different loss scale (e.g., BCE for classification vs. Dice for segmentation). Fixed weights (like 0.5 for each) can be suboptimal, especially when tasks learn at different speeds or have noisy labels. Overweighting one task can hurt the other's performance. Use task-dependent homoscedastic uncertainty to learn how much to trust each task. For each task t, introduce a learnable scalar σ_t representing how "uncertain" or noisy the task is.

Given an input $\mathbf{X}$ and a vector of task identifiers $\mathbf{t}$, the encoder outputs $\{\mathbf{F}_i\}$. Depending on the task type in $\mathbf{t}$, the corresponding head (segmentation or classification) is applied, producing $\hat{\mathbf{Y}}_{\text{seg}}$ or $\hat{\mathbf{y}}_{\text{cls}}$. The final model jointly optimizes both heads using a dynamic task-weighted loss:

$$\mathcal{L} = \sum_{k \in \mathcal{T}} \frac{1}{2\sigma_k^2} \mathcal{L}_k + \log \sigma_k,$$

where σ_k are learnable uncertainty parameters for each task k.

If the model can't reduce the loss much , it will increase σ_t and thus down-weight that task. If the model can optimize the task well, it reduces σ_t , effectively up-weighting it.

4 Experimental Results

Dataset: We use the same dataset as provided in the challenge [6]. And we follow the same data split as provided file.

Segmentation:

The DICE and IOU scores are used to evaluate the segmentation performance.

Classification: The dataset has different post-phases, but all subjects have at least three phases including the pre-contrast and two post-contrast phases.

To unify the input for the deep learning model, we retain the maximum average intensity value post-contrast phase t_m, along with the pre-contrast phase p_0 and the last post-contrast phase p_n, acquired as $[p_0, p_m, p_n]$. We use accuracy, area under the curve (AUC), F1 score, recall and Matthews correlation coefficient (MCC) to evaluate the classification performance.

The proposed method needs to train two models separately. In this first step, we only train the 2D feature extractor using the selected slice as input. In this second step, the transformer model is used to train on the extracted feature maps. The binary cross-entropy loss function was used to train these models.

Experiment setting: The segmentation experiments are conducted on a single NVIDIA RTX A100 GPU with 80GB of memory. We follow the same training procedure as the nnUNet model. The Adam optimizer was used for training with $3e^{-4}$ learning rate and 64 batch size.

4.1 Segmentation Result

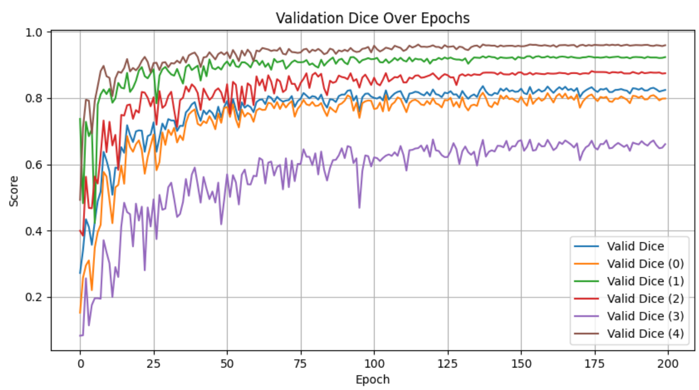

Fig. 2. The training procedure of different methods segmentation performance

As shown in Table 1, we evaluate the segmentation performance of different methods. The benchmark is denoted as the naive UNet model.

The Figure 2 is Dice score during the training procedure.

4.2 Classification Result

To compare the performance of different methods, we use various methods, including a separate UNet and classification model. The Figure 3 is AUC score during the training procedure.

Table 1. Comparison of the different Segmentation methods DICE score.

Methods	Breast	Cardiac	Thyroid	Kidney	Fetal↑
BenchMark	76.08	54.41	39.47	87.04	80.10
Unius	73.01	66.03	39.42	81.61	81.39
Ours	71.03	74.70	51.61	85.05	86.23

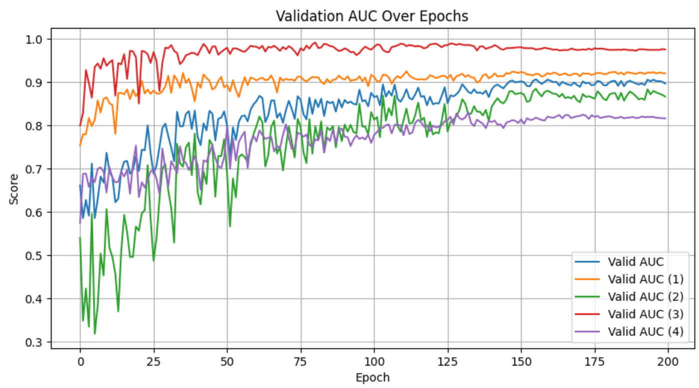

Fig. 3. The training procedure of different methods classification performance

Table 2. Comparison of the different classification methods AUC score.

Methods	Breast	Breast-S	Liver	Appendix ↑
BenchMark	76.08	54.41	39.47	87.04
Unius	50.00	50.00	50.00	50.00
Ours	78.45	51.93	86.46	83.54

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